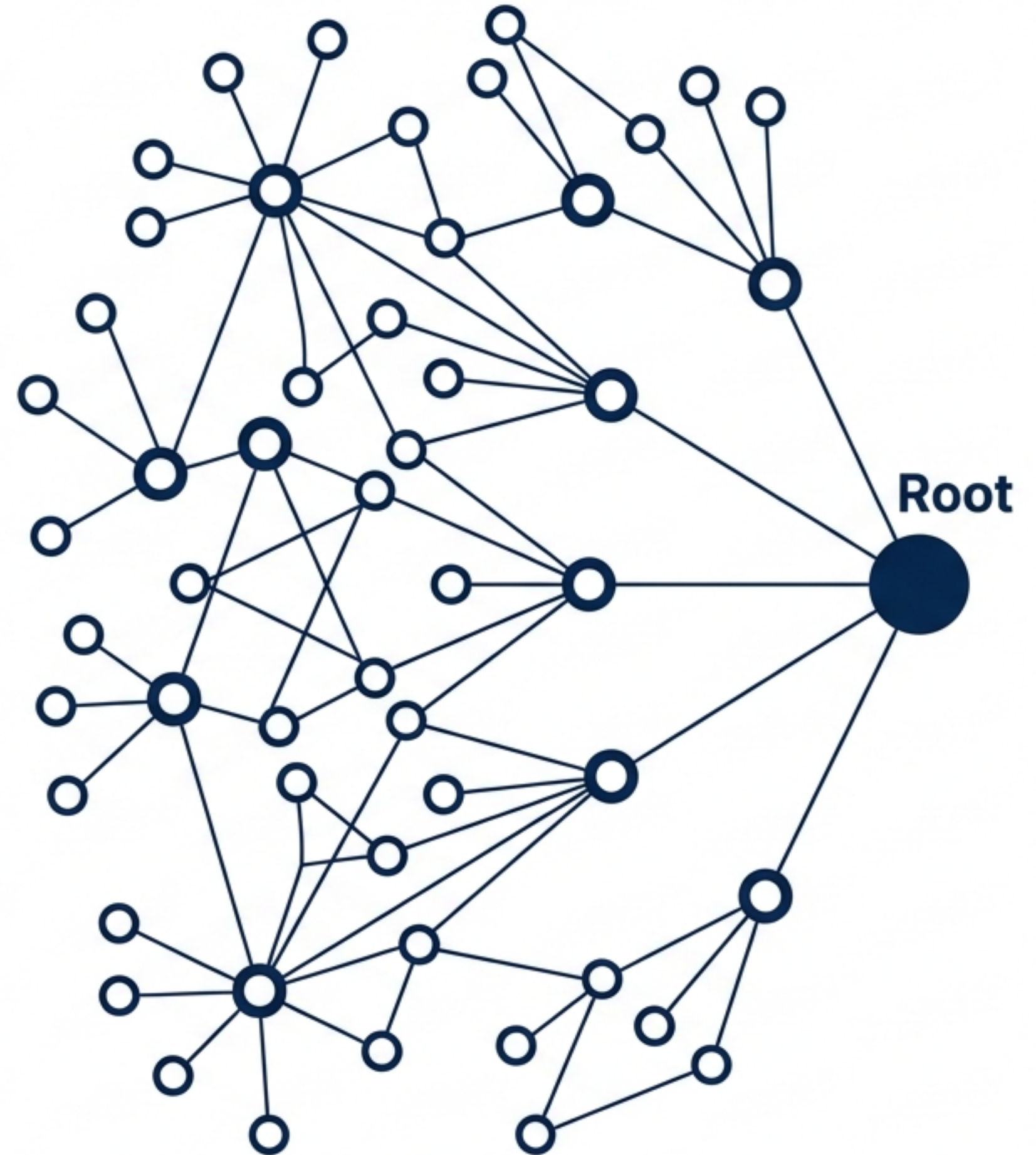


Demystifying the Global DNS

Architectural Topology &
Name Resolution Lab



Learning Objectives

- 1** Analyze the Global Namespace hierarchy, logical partitioning, and naming constraints.
- 2** Trace iterative and recursive resolution mechanisms across the system architecture.
- 3** Evaluate modern cryptographic integrations and transport security (DNSSEC, DoT, DoH).

Toolchain Setup



- **Target Interface:** Command Line (UNIX/Linux/macOS)
- **Resolution Query Tool:** dig (Domain Information Groper)
- **Traffic Analyzer:** Wireshark
- **Environment:** Unfiltered outbound UDP/TCP Port 53

```
$ dig localhost

; <>> DiG 9.10.6 <>> localhost
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 3951
;; flags: qr rd ra; QUERY: 1, ANSWER: 1

;; ANSWER SECTION:
localhost.      604800  IN      A       127.0.0.1
```

The Architecture of Identity

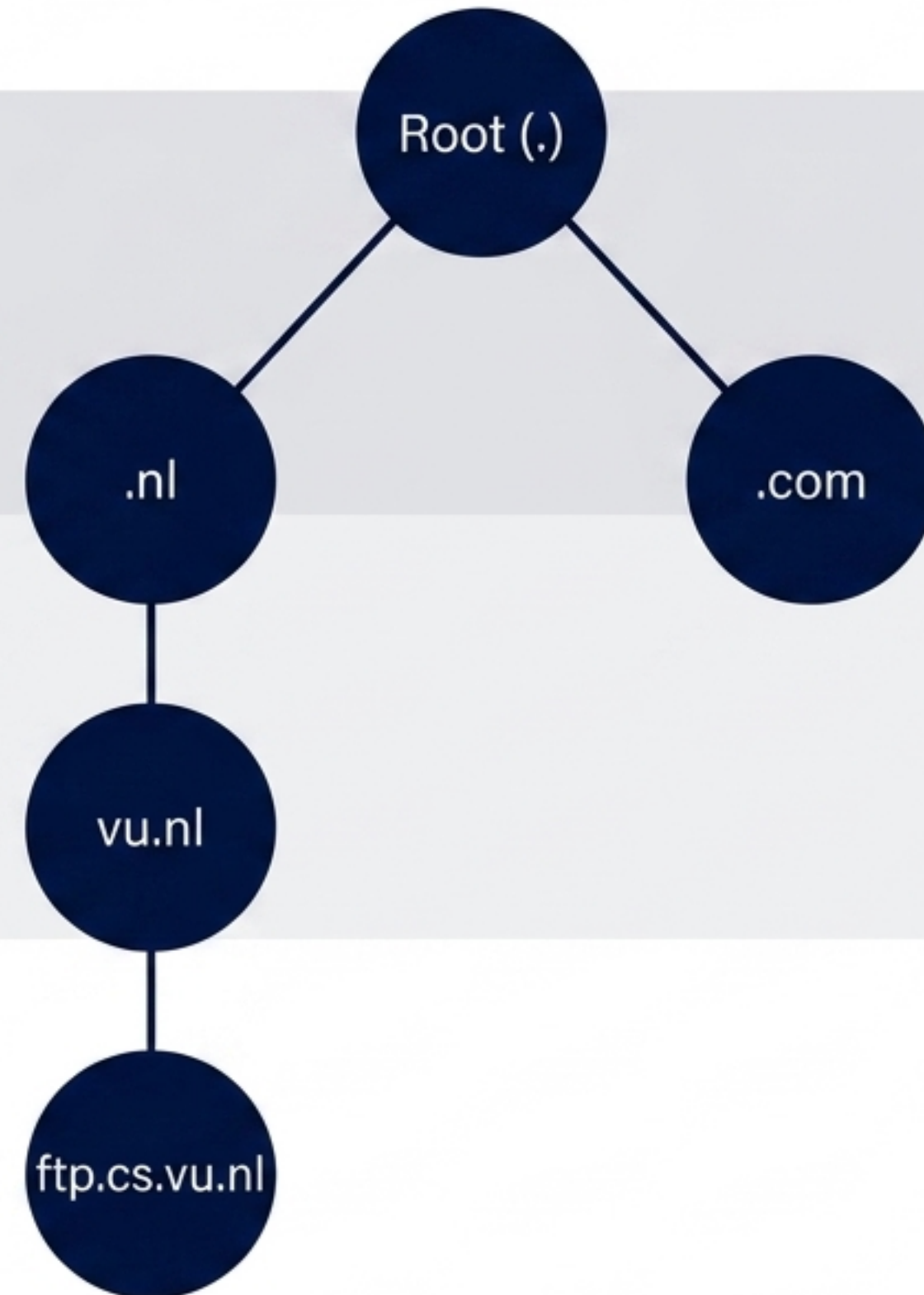
Feature	Flat Names	Structured Names
Form	Unstructured bit strings (e.g., Public Keys)	Hierarchical labels (max 63 chars per label, 255 total)
Machine Utility	Highly efficient for direct cryptographic verification	Enables logical partitioning and zoned delegation
Human Utility	Entirely unmemorable	Intuitive; maps exactly to logical data organization

Partitioning the Namespace

Global Layer: Root nodes and TLDs (.com, .nl).
Geographic scale: Worldwide.
Updates: Lazy.
Replication: Massive.

Administrational Layer: Organizations (vu.nl).
Geographic scale: Regional/Org.
Updates: Immediate.
Replication: Minimal.

Managerial Layer: Individual hosts (ftp.cs.vu.nl).
Geographic scale: Department.
Updates: Immediate.
Replication: None.



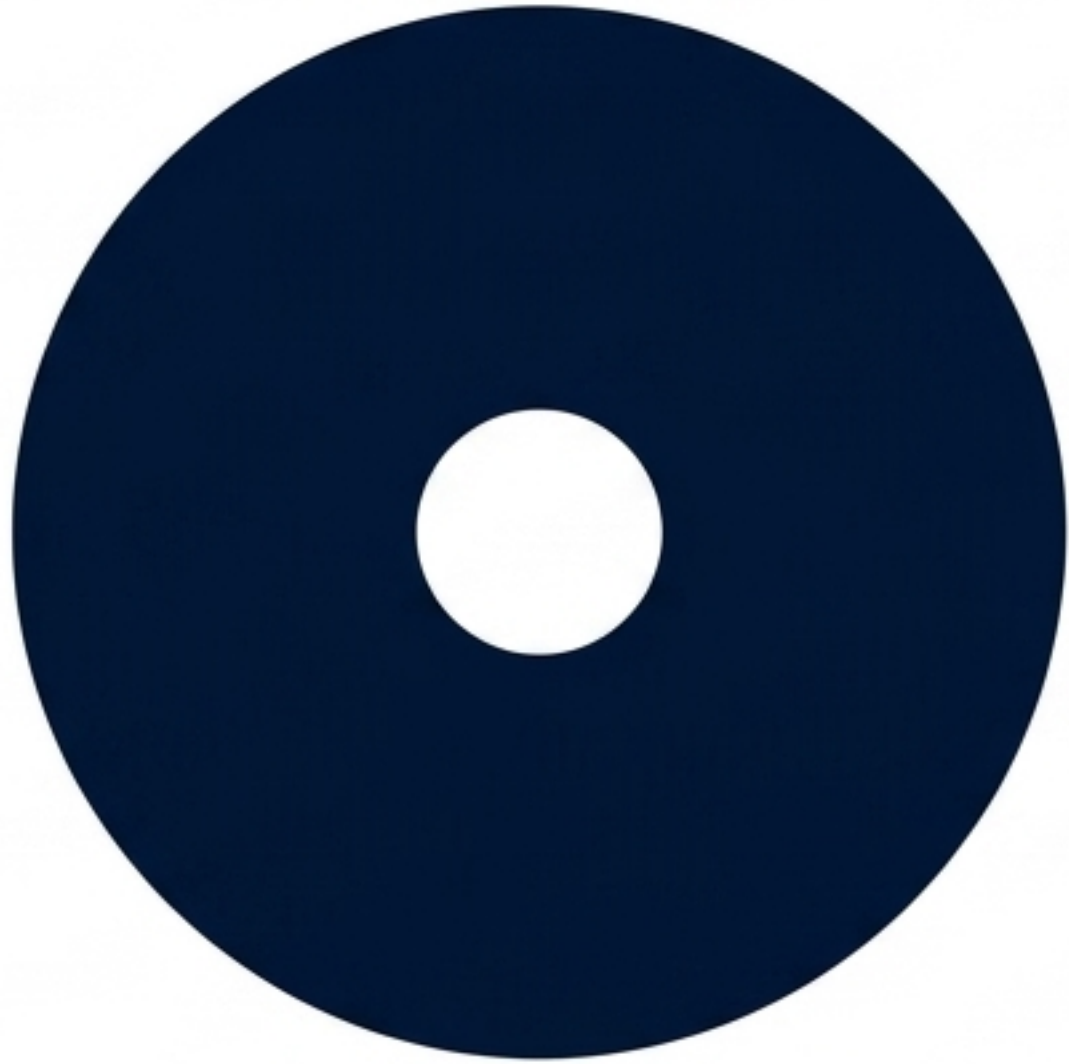
Step 1: Querying Resource Records

A / AAAA:	Maps hostname directly to IPv4/IPv6 routing address.
MX	Decouples web traffic from mail, routing to a specific mail handler.
NS	The glue of the hierarchy—delegates authoritative truth to a specific server.
CNAME	Provides a Canonical Name, allowing one physical server to hold multiple aliases.
SOA	Start of Authority containing technical parameters for zone maintenance.

```
$ dig vu.nl ANY

;; ANSWER SECTION:
vu.nl.      300 IN  A      130.37.20.20
vu.nl.      300 IN  MX     10 mail.vu.nl.
vu.nl.      3600 IN NS     ns1.vu.nl.
vu.nl.      3600 IN  SOA    ns1.vu.nl. admin.vu.nl.
```


Step 2: The Closure Mechanism



How does a system know where to start?

Every resolution requires an implicit, hard-coded starting point.

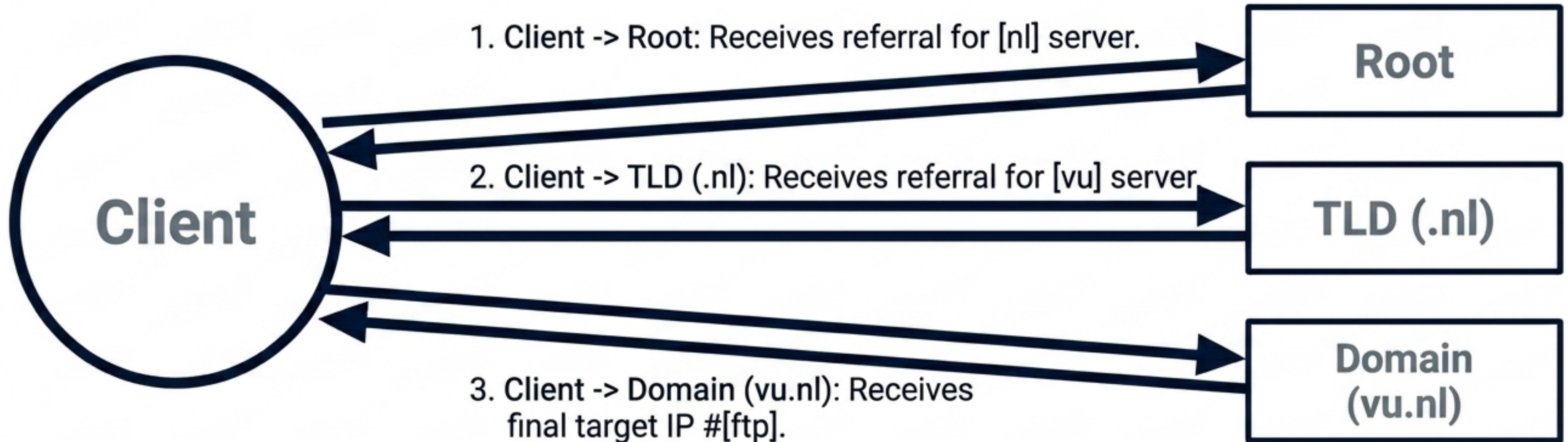
In UNIX, the closure mechanism is the superblock (calculating inode 0).

In DNS, the system relies on the root hints file to find the absolute anchor: the Root (.).

```
$ dig +trace .  
. 518400 IN NS a.root-servers.net.  
. 518400 IN NS b.root-servers.net.  
. 518400 IN NS c.root-servers.net.  
. 518400 IN NS d.root-servers.net.  
;; Received 239 bytes from 192.36.148.17#53(i.root-servers.net)
```


Step 3: Tracing Iterative Resolution

The Client-Coordinated Dialogue (e.g., root:[nl, vu, cs, ftp])



Key Constraint: Global root servers strictly enforce iterative resolution to prevent system collapse under massive load.

Step 4: Tracing Recursive Resolution

The Delegated Chain Reaction



1. Handoff: Client queries Local ISP Resolver.

2. Chain of Custody: Resolver manages the entire path navigation on the client's behalf.

3. Collective Intelligence: As the resolved IP travels back up the chain, intermediate servers cache the sub-tree mappings for future clients.

Architectural Trade-Offs

Metric	Iterative	Recursive
Server Load	Light (Referrals only)	Heavy (Manages active state)
Total Comm. Cost	High (e.g., 3 trans-Atlantic round trips)	Low (1 long trip, remaining lookups local)
System Caching	Poor (Client-side only)	Excellent (Shared intermediate ISP cache)



Core Insight: DNS enforces Iterative at the core for survival, and utilizes Recursive at the edge to exploit the Principle of Locality.

Step 5: Modern Security Protocols



DNSSEC (Security):
Cryptographic Trust Chain.
Parent zone signs the hash
of the Child's Key-Signing
Key (KSK) to validate the
Zone-Signing Keys (ZSK).
Prevents spoofing.



DNS over TLS (DoT):
Wraps standard queries in
TLS via Port 853.
Provides confidentiality
against network snoopers.



DNS over HTTPS (DoH):
Uses Port 443.
Bypasses local network
admin policy. Enhances
individual privacy but
disrupts organizational
traffic monitoring.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000	192.168.1.50	1.1.1.1	TLSv1.2	112	Application Data
2	0.015	1.1.1.1	192.168.1.50	TLSv1.2	164	Application Data
3	0.041	192.168.1.50	8.8.8.8	TCP	74	443 > 52142 [SYN, ACK]

Common Resolution Failures

Issue 1: Stale Cache Desynchronization

Symptom: Local resolver IP differs from the Authoritative server IP.

Root Cause: The Principle of Locality fails when Time-To-Live (TTL) decay values are set too high during a rapid Managerial-layer update.

Issue 2: Broken Trust Chain

Symptom: SERVFAIL response exclusively on DNSSEC-enabled zones.

Root Cause: Cryptographic mismatch; the Parent domain's DS record no longer matches the Child zone's KSK.

Assignment Deliverables



1. Resolution Trace Map: A documented terminal log mapping an iterative, step-by-step resolution path for a .nl domain.

2. Latency Analysis:

A statistical comparison report of recursive versus iterative communication costs based on your network location.

3. Packet Capture Log:

Exported Wireshark captures demonstrating the structural differences between plaintext UDP 53 and encrypted DoH traffic.



Advanced Architecture Challenges

Challenge A: Logical vs. Physical Routing

While there are only 13 logical root nodes, they are backed by immense physical replication. Map the Anycast routing of the RIPE NCC root node across its 25 physical sites.

Challenge B: The DHT Alternative

Contrast the hierarchical DNS caching model with a flat Distributed Hash Table (DHT) architecture. Why do DHTs struggle to match DNS's geographical efficiency?

“DNS remains the masterclass in distributed systems engineering because it successfully decouples a massive physical infrastructure from a simple logical hierarchy—scaling globally through the Principle of Locality.”